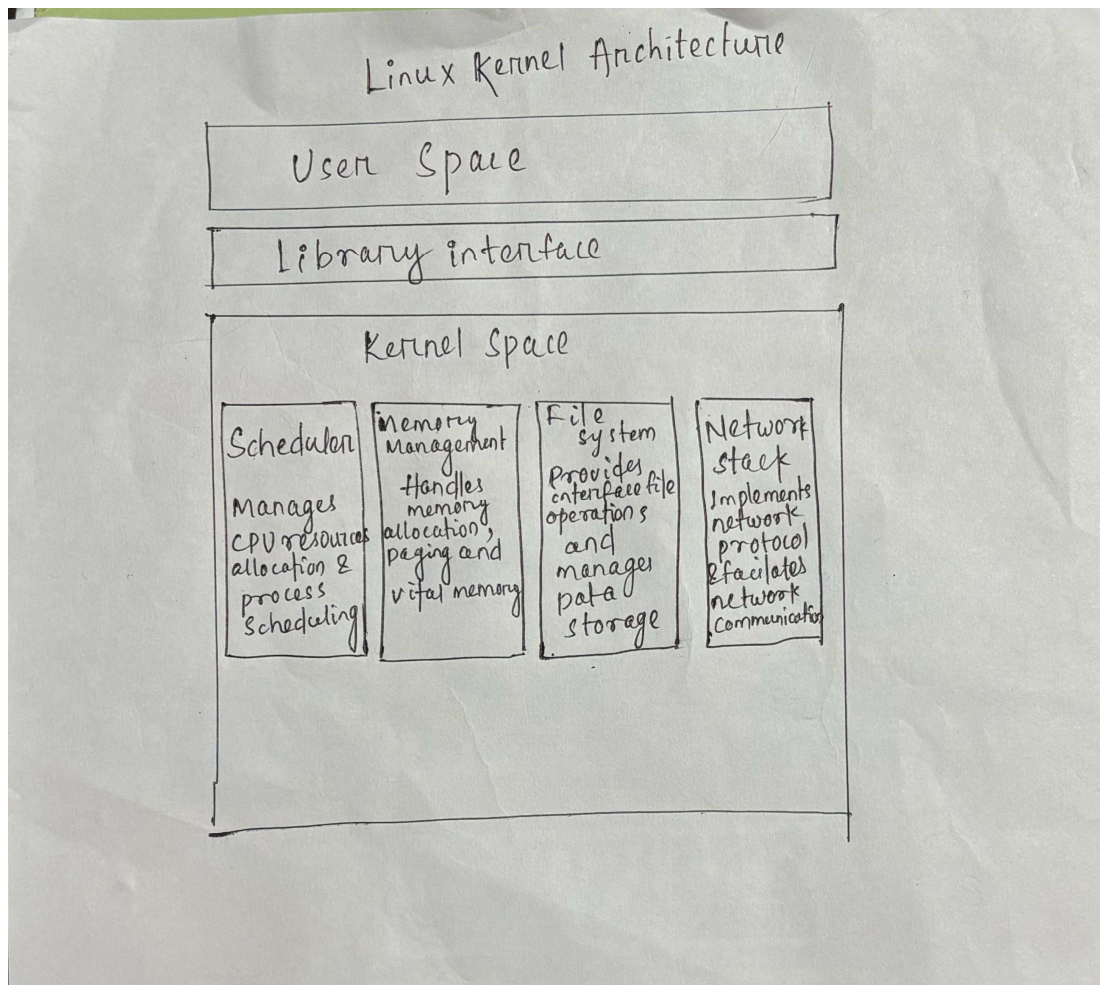


LINUX KERNEL ARCHITECTURE



Overview of Linux Kernel Architecture:

The Linux kernel is the core part of the operating system that manages hardware resources and system-level operations. Below is a concise explanation of its key components:

1. Process Scheduler:

The process scheduler coordinates the execution of processes by selecting which one should run at a given moment. It ensures balanced CPU usage across processes, utilizing a variety of scheduling strategies to support multitasking and responsiveness.

2. Memory Management Unit (MMU):

This unit oversees how memory is allocated to applications and reclaims it when no longer needed. It features support for virtual memory, paging, and memory protection to maintain optimal system performance and reliability.

3. File System:

The file system organizes data on storage media and offers mechanisms for data access and retrieval. It accommodates multiple formats such as ext4, FAT, and NTFS, and provides a consistent interface for file operations through system calls.

4. Networking Stack:

This component enables data exchange across networks by implementing standard protocols like TCP/IP. It is responsible for packet management, data integrity checks, and ensuring consistent communication between devices.

5. Inter-Process Communication (IPC):

IPC facilities, including shared memory, message queues, pipes, and semaphores, enable processes to coordinate and exchange data. These mechanisms are fundamental to supporting concurrent execution and complex workflows.

6. Device Drivers:

Device drivers serve as intermediaries between the hardware and the operating system. They provide standardized access to peripheral devices—such as storage drives, printers, and input devices—by translating OS commands into hardware-specific instructions.

7. System Call Interface (SCI):

The SCI acts as a gateway between user-level programs and kernel services. It converts application-level requests into kernel-level operations, allowing secure and controlled resource access.

8. Kernel and User Modes:

Linux enforces a mode-based execution model: kernel mode grants unrestricted system access for critical operations, while user mode restricts access to safeguard system integrity and prevent accidental damage.